

Surface Metrics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Text Descriptives on the Exchange-Aligned Structure

PENPAL Project

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1 Overview

This notebook compares surface-level text metrics across conditions using the long-format, exchange-aligned metric tables.

The underlying processed data are now one row per **speaker contribution**, with recovered role metadata:

- **Human / AI** in Human-AI
- **Starter-side / Non-starter-side** in Human-Human and AI-AI

Those role labels are useful for **within-condition description**, but they are not the primary cross-condition comparison axis. For cross-condition comparison, this notebook uses **Slot 1 / Slot 2 asymmetry**.

The notebook therefore no longer assumes old wide **author_1_*** / **author_2_*** columns.

Primary cross-condition object. Per story and per metric m ,

$$A_m(i) = \left| \overline{m}_{\text{slot}=1, i} - \overline{m}_{\text{slot}=2, i} \right|.$$

This is the unsigned, label-invariant side-difference magnitude. For each metric we test $A_m \sim$ condition with ANOVA + Kruskal and apply BH adjustment across the metric panel. The LMM slot-marginal contrasts are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

Article-use note. The primary result is the per-story unsigned side-difference test. Slot-marginal LMMs and role plots are useful for explaining where the difference comes from, but they should not replace the label-invariant cross-condition result.

2 Load Data

Table 1: Loaded surface-metric rows by condition

condition	n_stories	n_rows
Human-AI	100	1384
Human-Human	36	576
AI-AI	80	1280

3 Metric Selection

Table 2: Surface metrics selected for the comparison notebook

metric_alias	column_name
word_count	n_tokens
sentence_length	sentence_length_mean
lexical_diversity	proportion_unique_tokens
readability	flesch_reading_ease
coherence	first_order_coherence

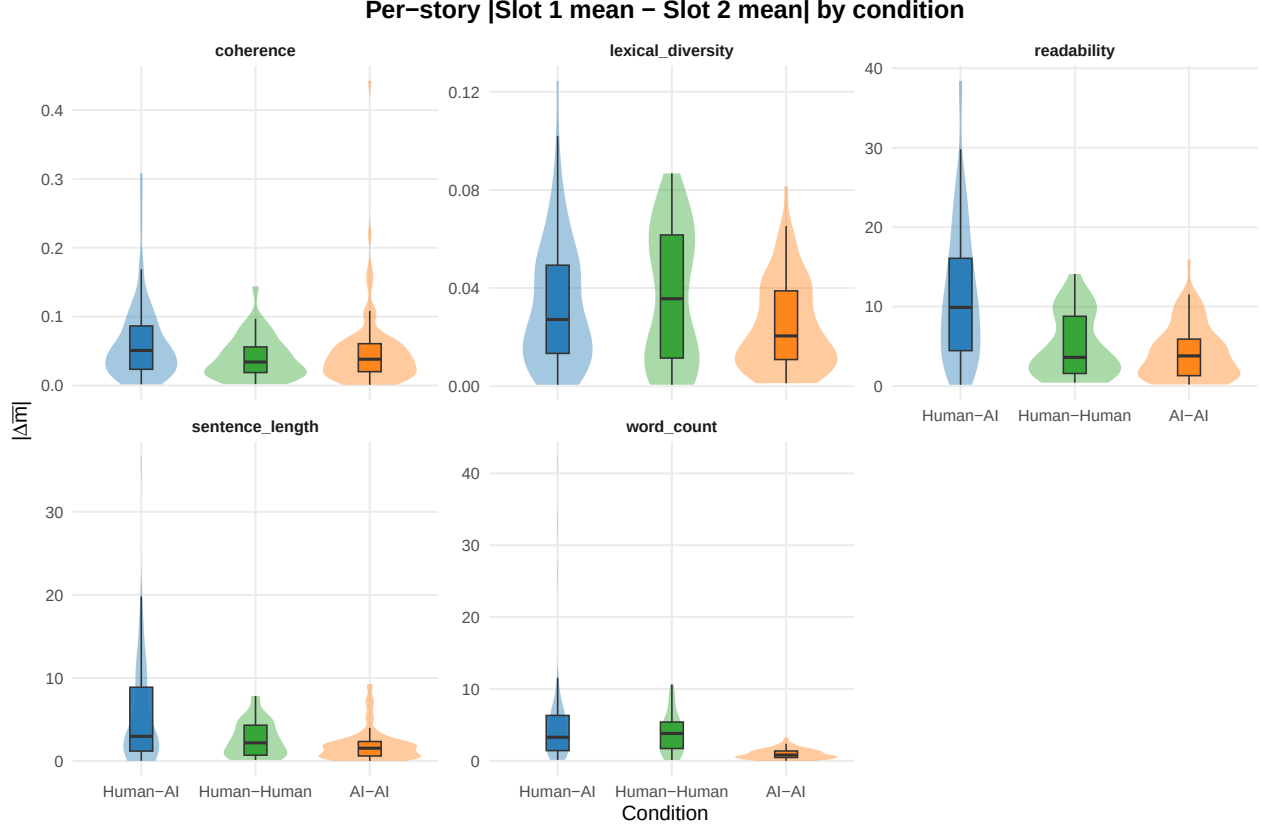
4 Primary: Per-Story Side-Mean Difference

Table 3: Per-story |side-mean difference| by condition, per metric

metric	condition	n	mean_A	median_A	sd_A
coherence	Human-AI	95	0.062	0.051	0.056
coherence	Human-Human	36	0.040	0.034	0.032
coherence	AI-AI	78	0.050	0.038	0.060
lexical_diversity	Human-AI	100	0.033	0.027	0.025
lexical_diversity	Human-Human	36	0.036	0.036	0.027
lexical_diversity	AI-AI	80	0.024	0.020	0.018
readability	Human-AI	100	11.494	9.904	8.627
readability	Human-Human	36	5.033	3.613	3.948
readability	AI-AI	80	4.284	3.797	3.422
sentence_length	Human-AI	100	5.447	2.980	5.930
sentence_length	Human-Human	36	2.636	2.191	2.120
sentence_length	AI-AI	80	1.945	1.547	1.966
word_count	Human-AI	100	4.829	3.286	5.935
word_count	Human-Human	36	3.806	3.812	2.752
word_count	AI-AI	80	0.955	0.812	0.709

Table 4: Primary cross-condition tests for A_m ~ condition, per metric (BH-adjusted across metrics)

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
coherence	0.0888	0.0446	0.0888	0.0462
lexical_diversity	0.0118	0.0462	0.0148	0.0462
readability	0.0000	0.0000	0.0000	0.0000
sentence_length	0.0000	0.0000	0.0000	0.0000
word_count	0.0000	0.0000	0.0000	0.0000



5 Bootstrap CIs and Effect Sizes per Metric

Table 5: Bootstrap 95% CI on per-condition mean of A per metric (R = 10,000)

metric	condition	n	mean	ci_lower	ci_upper
coherence	Human-AI	95	0.062	0.051	0.074
coherence	Human-Human	36	0.040	0.030	0.050
coherence	AI-AI	78	0.050	0.039	0.065
lexical_diversity	Human-AI	100	0.033	0.028	0.038
lexical_diversity	Human-Human	36	0.036	0.027	0.045
lexical_diversity	AI-AI	80	0.024	0.020	0.028
readability	Human-AI	100	11.494	9.893	13.236
readability	Human-Human	36	5.033	3.815	6.327

metric	condition	n	mean	ci_lower	ci_upper
readability	AI-AI	80	4.284	3.558	5.029
sentence_length	Human-AI	100	5.447	4.331	6.675
sentence_length	Human-Human	36	2.636	1.970	3.330
sentence_length	AI-AI	80	1.945	1.532	2.412
word_count	Human-AI	100	4.829	3.791	6.098
word_count	Human-Human	36	3.806	2.962	4.701
word_count	AI-AI	80	0.955	0.806	1.111

Table 6: Pairwise Cliff’s delta on A per metric

metric	contrast	n_a	n_b	cliff_delta	magnitude
coherence	Human-AI vs Human-Human	95	36	0.239	small
coherence	Human-AI vs AI-AI	95	78	0.175	small
coherence	Human-Human vs AI-AI	36	78	-0.071	negligible
lexical_diversity	Human-AI vs Human-Human	100	36	-0.065	negligible
lexical_diversity	Human-AI vs AI-AI	100	80	0.189	small
lexical_diversity	Human-Human vs AI-AI	36	80	0.224	small
readability	Human-AI vs Human-Human	100	36	0.487	large
readability	Human-AI vs AI-AI	100	80	0.557	large
readability	Human-Human vs AI-AI	36	80	0.106	negligible
sentence_length	Human-AI vs Human-Human	100	36	0.239	small
sentence_length	Human-AI vs AI-AI	100	80	0.393	medium
sentence_length	Human-Human vs AI-AI	36	80	0.220	small
word_count	Human-AI vs Human-Human	100	36	0.052	negligible
word_count	Human-AI vs AI-AI	100	80	0.697	large
word_count	Human-Human vs AI-AI	36	80	0.700	large

6 Planned Contrast: HA vs (HH + AA)/2 per Metric

Table 7: Planned HA vs (HH+AA)/2 contrast per metric, BH-adjusted across metrics

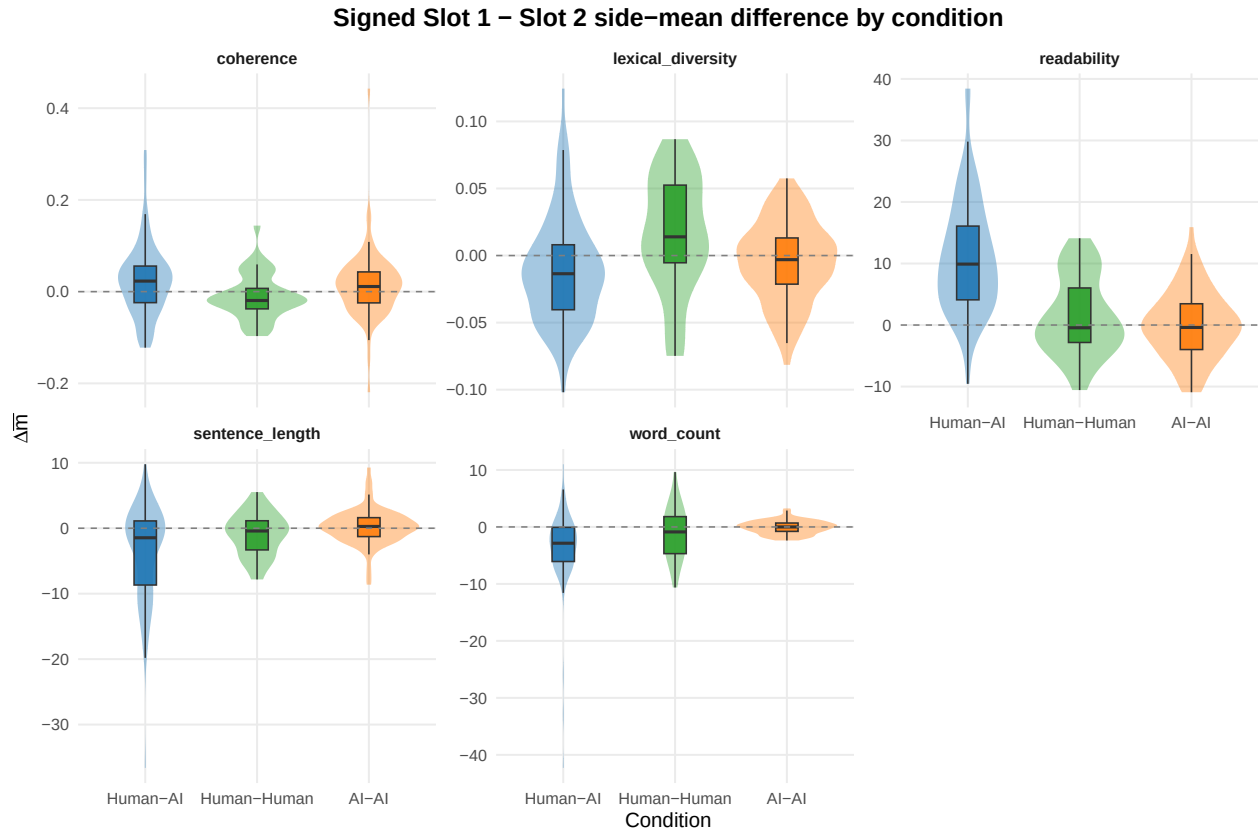
metric	contrast	estimate	se	t	df	p	ci_lower	ci_upper	p_BH
coherence	Human-AI vs (Human-Human + AI-AI)/2	0.017	0.007	2.367	180.328	0.019	0.003	0.031	0.024
lexical_diversity	Human-AI vs (Human-Human + AI-AI)/2	0.003	0.004	0.770	133.224	0.443	-0.004	0.010	0.443
readability	Human-AI vs (Human-Human + AI-AI)/2	6.836	0.943	7.250	132.924	0.000	4.971	8.701	0.000
sentence_length	Human-AI vs (Human-Human + AI-AI)/2	3.156	0.628	5.022	121.977	0.000	1.912	4.401	0.000
word_count	Human-AI vs (Human-Human + AI-AI)/2	2.449	0.638	3.841	123.969	0.000	1.187	3.711	0.000

7 Directional Stability of Signed Side-Mean Differences (per Metric)

Tests whether the *direction* of the slot-1 vs slot-2 side-mean difference is consistent across stories within each condition, per metric. The signed quantity is $\text{signed}_i = \bar{m}_{\text{slot } 1, i} - \bar{m}_{\text{slot } 2, i}$.

Table 8: Per-metric directional stability of signed side-mean differences

metric	condition	n	mean	sd	stability	sign_consistency	one_sample_tvariance_equality_p
coherence	Human-AI	95	0.025	0.080	0.312	0.642	0.003
coherence	Human-Human	36	-0.011	0.050	0.210	0.667	0.217
coherence	AI-AI	78	0.014	0.077	0.188	0.615	0.101
lexical_diversity	Human-AI	100	-0.011	0.040	0.262	0.670	0.010
lexical_diversity	Human-Human	36	0.015	0.043	0.351	0.694	0.043
lexical_diversity	AI-AI	80	-0.005	0.030	0.154	0.550	0.173
readability	Human-AI	100	11.020	9.231	1.194	0.940	0.000
readability	Human-Human	36	1.390	6.297	0.221	0.528	0.194
readability	AI-AI	80	0.024	5.504	0.004	0.525	0.969
sentence_length	Human-AI	100	-3.602	7.213	0.499	0.620	0.000
sentence_length	Human-Human	36	-0.938	3.276	0.286	0.583	0.095
sentence_length	AI-AI	80	0.230	2.765	0.083	0.575	0.459
word_count	Human-AI	100	-3.551	6.786	0.523	0.750	0.000
word_count	Human-Human	36	-1.014	4.627	0.219	0.583	0.197
word_count	AI-AI	80	-0.052	1.193	0.043	0.488	0.700

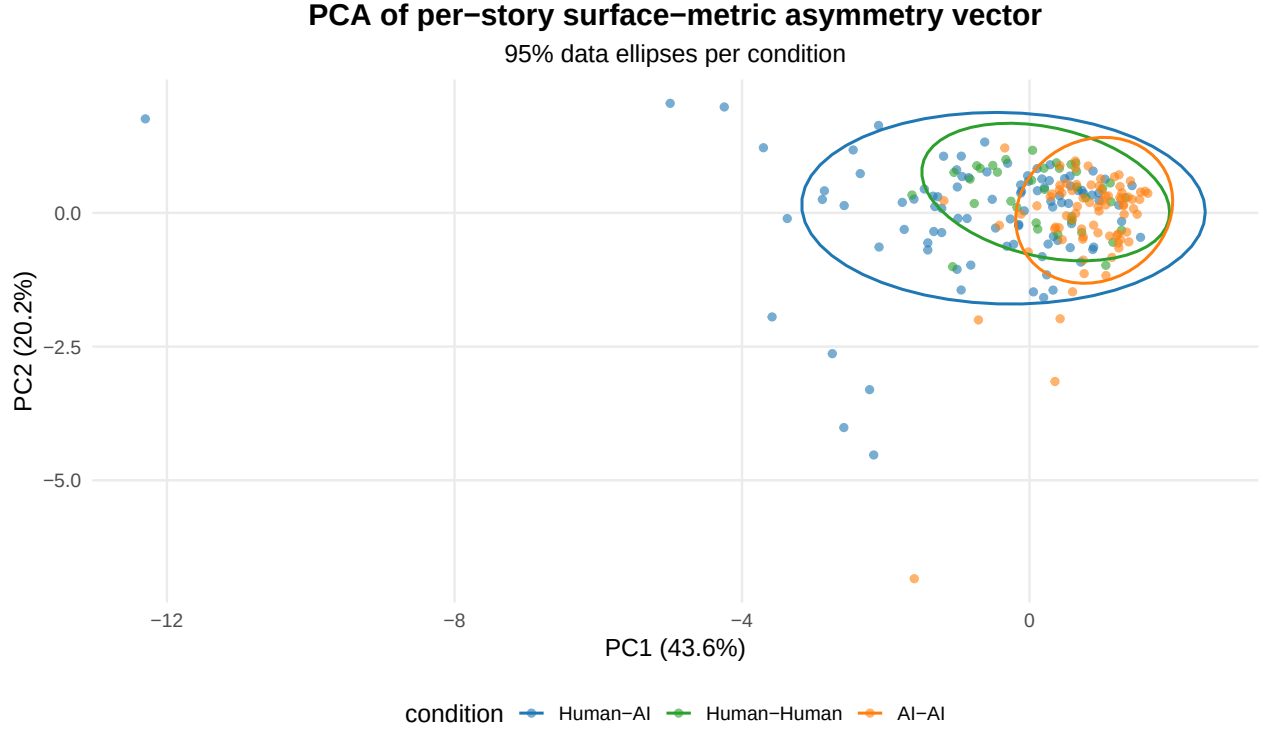


8 Multivariate Composite over Surface Metrics

Stacks the per-story A_m vector across all selected surface metrics into one matrix and tests whether the joint condition profiles differ via PERMANOVA (`vegan::adonis2`). A single multivariate answer to “is the cross-metric surface-asymmetry profile distinct in HA?”.

```
## Stories with complete A-vector across all 5 metrics: 209

## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## vegan::adonis2(formula = X ~ condition, data = A_matrix, permutations = 9999, method = "euclidean")
##           Df SumOfSqs      R2      F Pr(>F)
## Model      2   3281.2 0.17854 22.387 1e-04 ***
## Residual 206  15096.5 0.82146
## Total    208  18377.7 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

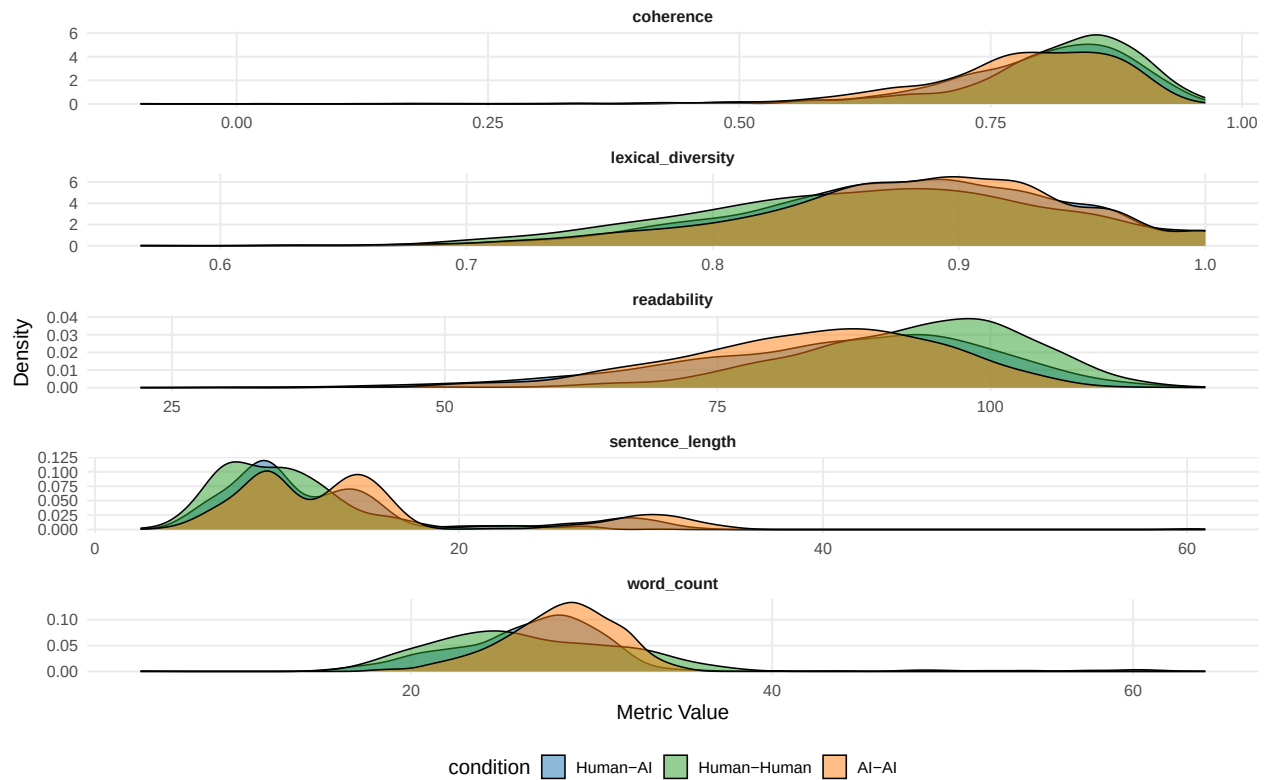


9 Supplemental: General Condition-Level Comparison

Table 9: Condition-level descriptives for selected surface metrics

condition	metric	mean_value	sd_value
Human-AI	word_count	27.412	6.409
Human-Human	word_count	26.365	4.911
AI-AI	word_count	28.263	3.109
Human-AI	sentence_length	13.147	7.152
Human-Human	sentence_length	10.533	4.253
AI-AI	sentence_length	14.639	7.553
Human-AI	lexical_diversity	0.879	0.064
Human-Human	lexical_diversity	0.866	0.071
AI-AI	lexical_diversity	0.881	0.065
Human-AI	readability	85.349	14.439
Human-Human	readability	92.835	10.933
AI-AI	readability	83.504	12.043
Human-AI	coherence	0.797	0.101
Human-Human	coherence	0.818	0.085
AI-AI	coherence	0.780	0.103

Surface Metric Distributions by Condition



```
##
## === word_count ( n_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 18544.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.8520 -0.5427  0.0354  0.5580  5.9413
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 10.99      3.316
## Residual                15.21      3.901
## Number of obs: 3240, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    27.3915    0.3478 214.3884  78.761  <2e-16 ***
## conditionHuman-Human -1.0269    0.6729 210.4826  -1.526    0.128
## conditionAI-AI      0.8718    0.5199 211.4343   1.677    0.095 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```



```

##          (Intr) cndH-H
## cndtnHmn-Hm -0.517
## condtnAI-AI -0.669  0.346

## contrast          estimate    SE  df z.ratio p.value
## (Human-AI) - (Human-Human)    1.027 0.673 Inf   1.526  0.3809
## (Human-AI) - (AI-AI)         -0.872 0.520 Inf  -1.677  0.2807
## (Human-Human) - (AI-AI)      -1.899 0.694 Inf  -2.737  0.0186
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === sentence_length ( sentence_length_mean ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 20480.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9578 -0.5648 -0.1443  0.4738  6.3716
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 19.95     4.467
## Residual                27.67     5.260
## Number of obs: 3240, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    13.1500    0.4686 216.4286  28.065  <2e-16 ***
## conditionHuman-Human -2.6166    0.9065 212.4809  -2.886   0.0043 **
## conditionAI-AI      1.4891    0.7004 213.4428   2.126   0.0346 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) cndH-H
## cndtnHmn-Hm -0.517
## condtnAI-AI -0.669  0.346

## contrast          estimate    SE  df z.ratio p.value
## (Human-AI) - (Human-Human)    2.62 0.907 Inf   2.886  0.0117
## (Human-AI) - (AI-AI)         -1.49 0.700 Inf  -2.126  0.1005
## (Human-Human) - (AI-AI)      -4.11 0.934 Inf  -4.394 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === lexical_diversity ( proportion_unique_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))

```

```

## Data: df_combined
##
## REML criterion at convergence: -8713.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.6893 -0.6136  0.0658  0.6820  2.5374
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.000703 0.02651
## Residual                    0.003603 0.06002
## Number of obs: 3240, groups: conversation_id, 216
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.879021   0.003104 223.335675 283.148 <2e-16 ***
## conditionHuman-Human -0.013023   0.005952 211.448614  -2.188  0.0297 *
## conditionAI-AI       0.001888   0.004609 214.326820   0.410  0.6824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.522
## condtnAI-AI -0.674  0.351
##
## contrast            estimate      SE df z.ratio p.value
## (Human-AI) - (Human-Human)  0.01302 0.00595 Inf   2.188  0.0860
## (Human-AI) - (AI-AI)       -0.00189 0.00461 Inf  -0.410  1.0000
## (Human-Human) - (AI-AI)    -0.01491 0.00611 Inf  -2.439  0.0442
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === readability ( flesch_reading_ease ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 25263.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4674 -0.6088  0.0803  0.6701  2.9277
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 41.54      6.445
## Residual                    126.93    11.266
## Number of obs: 3240, groups: conversation_id, 216
##
## Fixed effects:

```

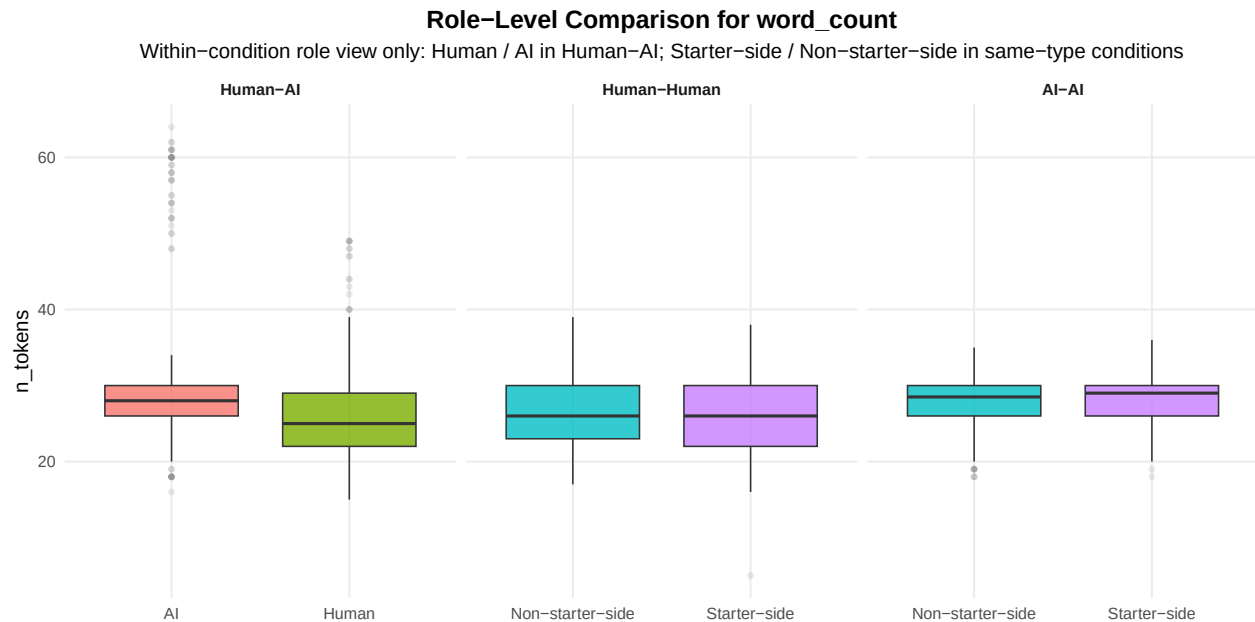
```

##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      85.3283    0.7123 218.1352 119.799 < 2e-16 ***
## conditionHuman-Human  7.5070    1.3717 210.2974   5.473 1.26e-07 ***
## conditionAI-AI      -1.8243    1.0610 212.2011  -1.719   0.087 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.519
## condtnAI-AI -0.671  0.349
##
## contrast      estimate    SE  df z.ratio p.value
## (Human-AI) - (Human-Human)  -7.51 1.37 Inf  -5.473 <.0001
## (Human-AI) - (AI-AI)         1.82 1.06 Inf   1.719 0.2566
## (Human-Human) - (AI-AI)      9.33 1.41 Inf   6.610 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === coherence ( first_order_coherence ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: -5020.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.7405 -0.4519  0.1673  0.6480  1.9687
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.0007897 0.02810
## Residual                0.0090040 0.09489
## Number of obs: 2780, groups: conversation_id, 216
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.797455    0.003982 222.324184 200.240 < 2e-16 ***
## conditionHuman-Human  0.020358    0.007374 188.682298   2.761 0.00633 **
## conditionAI-AI      -0.017091    0.005915 205.358893  -2.889 0.00427 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.540
## condtnAI-AI -0.673  0.364
##
## contrast      estimate    SE  df t.ratio p.value
## (Human-AI) - (Human-Human)  -0.0204 0.00737 192  -2.761 0.0190
## (Human-AI) - (AI-AI)         0.0171 0.00592 209   2.889 0.0128
## (Human-Human) - (AI-AI)      0.0374 0.00759 186   4.932 <.0001

```

```
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

10 Supplemental: Role-Level Comparison



```
##
## === Role contrast for word_count (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 3361.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.0125 -0.7251 -0.0490  0.7082  2.7514
##
## Random effects:
##  Groups      Name      Variance Std.Dev.
## conversation_id (Intercept)  6.244   2.499
## Residual                17.888   4.229
## Number of obs: 576, groups: conversation_id, 36
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    26.7917    0.4853  46.3676  55.201  <2e-16 ***
## condition_roleStarter-side  -0.8542    0.3525  539.0000  -2.424   0.0157 *
```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cndtn_rlSt- -0.363
## contrast                estimate    SE  df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.854 0.352 539   2.424  0.0157
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 6172.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6661 -0.6029  0.0512  0.6761  2.8658
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept)  3.399      1.844
## Residual                6.310      2.512
## Number of obs: 1280, groups:  conversation_id, 80
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      2.826e+01  2.288e-01 9.621e+01 123.525  <2e-16 ***
## condition_roleStarter-side  4.688e-03  1.404e-01 1.199e+03   0.033   0.973
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cndtn_rlSt- -0.307
## contrast                estimate    SE  df t.ratio p.value
## (Non-starter-side) - (Starter-side) -0.00469 0.14 1199  -0.033  0.9734
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for sentence_length (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 3294.6

```

```

##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.7213 -0.6670 -0.1741  0.3979  4.6936
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept)  0.9313   0.965
## Residual                    17.1848   4.145
## Number of obs: 576, groups: conversation_id, 36
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)         10.6959      0.2925  81.0235  36.571  <2e-16 ***
## condition_roleStarter-side -0.3251      0.3455 539.0000  -0.941   0.347
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.591
## contrast                estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.325 0.345 539   0.941  0.3471
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##      Data: df_subset
##
## REML criterion at convergence: 7904.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1311 -0.5805 -0.0766  0.6049  3.9973
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 34.40      5.865
## Residual                    23.07      4.803
## Number of obs: 1280, groups: conversation_id, 80
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)         14.6938      0.6827  85.4742  21.524  <2e-16 ***
## condition_roleStarter-side -0.1094      0.2685 1199.0000  -0.407   0.684
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)

```

```

## cndtn_rlSt- -0.197
## contrast estimate SE df t.ratio p.value
## (Non-starter-side) - (Starter-side) 0.109 0.268 1199 0.407 0.6837
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for lexical_diversity (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: -1411.9
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.2184 -0.6593 0.0593 0.7038 2.1836
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.0004991 0.02234
## Residual 0.0046142 0.06793
## Number of obs: 576, groups: conversation_id, 36
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 8.637e-01 5.467e-03 6.476e+01 157.987 <2e-16 ***
## condition_roleStarter-side 4.627e-03 5.661e-03 5.390e+02 0.817 0.414
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## cndtn_rlSt- -0.518
## contrast estimate SE df t.ratio p.value
## (Non-starter-side) - (Starter-side) -0.00463 0.00566 539 -0.817 0.4140
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: -3596.7
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.3098 -0.6313 0.0765 0.6719 2.7891

```

```
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.001108 0.03329
## Residual              0.003088 0.05557
## Number of obs: 1280, groups: conversation_id, 80
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.814e-01  4.322e-03 1.040e+02 203.938  <2e-16
## condition_roleStarter-side -9.556e-04  3.106e-03 1.199e+03  -0.308   0.758
##
## (Intercept)          ***
## condition_roleStarter-side
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.359
## contrast              estimate      SE    df t.ratio p.value
## (Non-starter-side) - (Starter-side) 0.000956 0.00311 1199   0.308  0.7584
##
## Degrees-of-freedom method: kenward-roger
```

11 Summary

Primary cross-condition result: per-metric $A_m(i) = |\overline{m}_{\text{slot } 1,i} - \overline{m}_{\text{slot } 2,i}|$, tested with ANOVA + Kruskal and BH-adjusted across metrics.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_m , per metric
- pairwise Cliff's δ effect sizes per metric
- planned single-df contrast HA vs (HH+AA)/2 per metric, BH-adjusted across metrics
- multivariate composite over the surface-metric panel via PERMANOVA + PCA biplot

Supplemental views retained:

- condition-level distributions and marginal `condition` LMMs
- within-condition role comparisons restricted to HH / AA (Starter-side / Non-starter-side); HA Human-vs-AI role LMM lives in `analysis/human-ai/surface_metrics_analysis.Rmd` and is not duplicated here

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized side differences across all conditions)

Analysis completed: 2026-05-05 12:46:50.894225